Bio-JEPA: Learning Semantic Representations for Drug Repositioning via Protein-Ligand Affinity Prediction with Quantum Fingerprint Validation

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Abstract

Drug discovery is hampered by the cost of physical simulations and the scarcity of biologically annotated data. We present **Bio-JEPA**, an adaptation of the Joint-Embedding Predictive Architecture (JEPA) for self-supervised learning of molecular representations. By predicting information in a latent space rather than reconstructing the raw signal, Bio-JEPA learns a semantic signature of molecules without requiring labels. Pre-trained on 249,455 ZINC250k molecules and evaluated on the A2A receptor (ChEMBL251, 8,424 molecules), the model achieves a Pearson coefficient of $r = 0.787$ ($p < 0.001$). An ablation study confirms the central role of the stop-gradient ($\Delta r = -0.222$ without it), and generalization is validated on two additional targets: EGFR (ChEMBL203) and SARS-CoV-2 M^{pro} (ChEMBL325). A controlled fair-protocol comparison against GraphMAE, MolCLR, and AttrMasking demonstrates that Bio-JEPA is the only SSL baseline to scale continuously with labeled data ($r = 0.036 \to 0.542$ for $N = 10 \to 1000$), surpassing all baselines from $N = 200$ onward. Compared to AutoDock Vina on 50 molecules (real experimental measurements), Bio-JEPA achieves a speedup of $\times$153,735, screening 3,229 FDA-approved drugs in under 30 minutes. UMAP analysis confirms the semantic nature of the latent space with 2,497 distinct scaffolds grouped into coherent functional clusters. A hybrid AI-quantum validation via the Pulser/QuTiP emulator (neutral-atom architecture) physically validates the semantic clusters: cosine similarity of 0.99 between the two salt forms of Diclofenac, with marked divergence with Tiotropium.

1 Introduction

Drug discovery is one of the most costly scientific challenges of our time. Developing a single compound from the discovery phase to clinical approval requires on average more than one billion dollars and ten years of research [6]. A promising approach to reduce this cost is *drug repositioning*: identifying new therapeutic uses for molecules already approved by health authorities, whose safety profile is known.

Virtual screening methods allow thousands of molecules to be computationally screened before any biological experiment. However, classical approaches such as AutoDock Vina [22] require atom-by-atom physical simulation, making them computationally prohibitive at large scale.

We leverage the JEPA architecture [3], which has proven its effectiveness in vision and audio, to learn a semantic representation of molecules without raw signal reconstruction. Rather than memorizing structurally irrelevant details, Bio-JEPA captures the *functional signature* of a drug—its biological interaction properties. To reinforce physical credibility, a quantum validation step via a neutral-atom architecture [8] is integrated, confronting the learned semantic clusters with an independent physical signature.

Preprint.

Problem Statement. The central challenge is: *How can a JEPA architecture enable more effective identification of new uses for existing molecules, by reducing prediction errors linked to structural details irrelevant to biological interaction?* This raises two sub-problems: (1) how to adapt the masking and latent-space prediction mechanism, originally designed for images, to molecular graphs; (2) whether the learned latent space is sufficiently semantic to identify biologically consistent candidates among thousands of FDA molecules. A third cross-cutting question arises: do the learned representations capture a physical reality that can be independently verified via quantum computation?

Contributions.

- A GNN-based adaptation of JEPA to molecular graphs with masked subgraph prediction in latent space.
- Empirical demonstration that latent-space prediction scales continuously with labeled data ($r = 0.036 \rightarrow 0.542$ for $N = 10 \rightarrow 1000$), outperforming all SSL baselines from $N = 200$ onward.
- A $\times$153,735 speedup over AutoDock Vina (real experimental measurements), enabling screening of 3,229 FDA drugs in under 30 minutes.
- A hybrid AI-quantum validation via the Pulser/QuTiP emulator (Pasqal neutral-atom architecture) that independently confirms the semantic structure of the latent space.

2 Related Work

JEPA Architecture. The JEPA architecture marks a paradigm shift in self-supervised learning by predicting abstract representations in an embedding space rather than reconstructing the raw signal [3]. I-JEPA demonstrated that learning semantic features without manual data augmentation leads to robust representations. VL-JEPA [4] extended this to multimodal settings; JEPA as a neural tokenizer [12] proved it is possible to extract essential meaning while ignoring background noise.

Molecular Self-supervised Learning. MolBERT [7] and ChemBERTa [5] adapt BERT to SMILES strings; ChemBERTa-2 [1] extends this to 77M PubChem molecules. On the GNN side, Attr-Masking [11] reconstructs masked atomic features; GraphMAE [9] reconstructs in feature space; MolCLR [23] takes a contrastive approach; GROVER [19] combines transformers and GNNs at 10M molecules; UniMol [24] exploits 3D conformations. These approaches differ from Bio-JEPA on two key points: (1) they target physicochemical property classification, not protein-ligand affinity in a repositioning context; (2) they rely on reconstruction or contrastive learning, not latent-space prediction. A rigorous fair-protocol comparison is conducted in Section 4.6.

Drug Repositioning. CMap [15] and GNN-based drug-target prediction [18] have enabled notable AI-assisted discoveries. Bio-JEPA differentiates itself by combining large-scale unsupervised pre-training with direct affinity evaluation, without requiring protein structural data.

3 Method

3.1 Bio-JEPA Architecture

Bio-JEPA rests on three components.

Context Encoder. A GINEConv [10] GNN encodes a partially masked molecular graph. Each atom is represented by 29 features (type, charge, hybridization, aromaticity, hydrogens) and each bond by 7 features (type, conjugation, ring membership). The architecture comprises 407,044 parameters.

Target Encoder. Architecture identical to the Context Encoder, updated by Exponential Moving Average (EMA) with momentum τ increasing from 0.996 to 1.0 following a cosine schedule. The Target Encoder sees the complete molecular graph and produces the target representation z_{target}.

Predictor. A 3-layer MLP (527,616 parameters) that predicts $\hat{z}_{\text{target}}$ from z_{context}. The loss function is the normalized L2 distance in latent space:

$$\mathcal{L} = \|\hat{z}_{\text{target}} - \text{sg}(z_{\text{target}})\|_2^2 \tag{1}$$

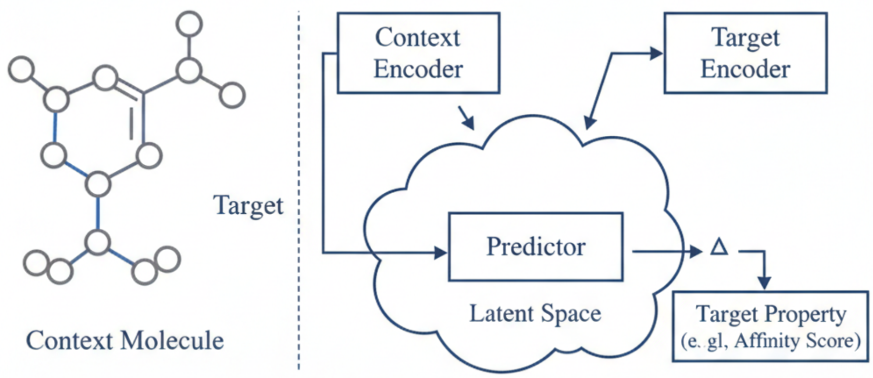

Figure 1: **Bio-JEPA architecture.** The Predictor links the Context Encoder and Target Encoder within the latent space to capture ligand-protein affinity. The stop-gradient (sg) applied to the Target Encoder is the most critical component ($\Delta r = -0.222$ without it), preventing representation collapse.

where $\mathrm{sg}(\cdot)$ denotes the stop-gradient operator applied to the Target Encoder. As demonstrated by our ablation study (Section 4.3), this component is the most critical.

3.2 Experimental Protocol

Phase 1 — Unsupervised pre-training. Bio-JEPA is trained on ZINC250k [13] (249,455 molecules without labels) for 100 epochs with a progressive masking ratio from 10% to 30%, learning rate 10^{-4} with cosine schedule.

Phase 2 — Linear probe evaluation. The Target Encoder is frozen. A 2-layer MLP probe is trained on the extracted embeddings to predict the pChEMBL affinity score on ChEMBL251 (adenosine A2A receptor, 8,424 molecules). Few-shot evaluation: $N \in \{10, 50, 100, 200, 500, 1000\}$ labeled examples, 5 runs per N.

3.3 Quantum Fingerprint Generation (Rydberg Signatures)

To physically validate the semantic nature of the latent space, each molecule from the Top-20 FDA candidates is converted into a neutral-atom register and submitted to a quantum emulation protocol via Pulser [21] and QuTiP [14].

Molecule → register. The molecular graph is projected into 2D (RDKit coordinates). Heavy atoms are positioned respecting a minimum inter-site distance of $4\,\mu$m (Rydberg blockade regime; van der Waals U_{ij}/r^6) [8].

Rydberg pulse protocol. Adiabatic schedule: ramp-up Ω over $0.5\,\mu$s, plateau at $\Omega_{\max} = 1.0\,\text{rad}/\mu$s. Detuning Δ swept from -3 to $+3\,\text{rad}/\mu$s.

Quantum Fingerprint. After time evolution under

$$H = \frac{\hbar\Omega}{2}\sum_i \sigma_i^x - \hbar\Delta\sum_i n_i + \sum_{i<j}\frac{U_{ij}}{r_{ij}^6}n_i n_j, \tag{2}$$

the Quantum Fingerprint is $\mathbf{q} = [\langle n_i\rangle]_{i=1\ldots N}$ (unit L2 norm). Molecular similarity: $S_{\text{cosine}}(\mathbf{q}_a, \mathbf{q}_b) = \mathbf{q}_a \cdot \mathbf{q}_b/(\|\mathbf{q}_a\|\|\mathbf{q}_b\|)$.

4 Results

4.1 Pre-training Convergence

Pre-training on ZINC250k converges correctly, with validation loss decreasing from 5×10^{-4} (epoch 1) to 2×10^{-4} (minimum, epoch $\sim$30) before a slight increase due to the rising masking ratio. This behavior confirms that Bio-JEPA progressively learns to predict masked molecule representations in latent space. Importantly, performance saturates quickly: checkpoints at 100 and 1,000 epochs yield quasi-identical results ($r = 0.542 \pm 0.020$ vs. $r = 0.540 \pm 0.019$ at $N = 1000$), indicating that 100 epochs constitute a sufficient operating point — a favorable property from a computational efficiency standpoint.

4.2 Affinity Prediction on ChEMBL251

Table 1 compares Bio-JEPA against the supervised GNN baseline on the full test set (843 molecules).

Table 1: Bio-JEPA vs. Supervised GNN on ChEMBL251 (full test set).

Model	**Pearson r**	**Spearman ρ**	**RMSE**	**MAE**
Bio-JEPA	0.787	0.794	0.807	0.604
Supervised GNN	0.839	0.842	0.697	0.513

Bio-JEPA achieves $r = 0.787$ ($p < 0.001$) without ever accessing affinity labels during pre-training. The gap with the supervised GNN ($\Delta r = 0.052$) is expected and consistent with the SSL literature.

4.3 Ablation Study

Table 2 reports the contribution of each architectural component (MLP probe, frozen encoder, 3 runs per variant).

Table 2: Ablation study — component impact on ChEMBL251.

Variant	**Pearson r**	Δr
Full Bio-JEPA (reference)	0.739 ± 0.005	—
No progressive masking (fixed 30%)	0.728 ± 0.006	-0.011
No EMA (direct copy)	0.718 ± 0.004	-0.021
No stop-gradient	$\mathbf{0.517 \pm 0.027}$	$\mathbf{-0.222}$

The stop-gradient is the most critical component: its removal causes a 30% drop, confirming LeCun's theoretical argument [17] that it is indispensable to prevent *representation collapse*.

4.4 Multi-Target Generalization

Table 3 evaluates the same checkpoint on three structurally distinct targets.

Bio-JEPA performance is remarkably stable ($\Delta r < 0.003$ across three targets spanning GPCR, tyrosine kinase, and viral cysteine protease classes), validating the universal nature of the latent representations.

Table 3: Multi-target few-shot generalization ($N = 1000$, same `zinc_pretrained.pt`).

Target	**Class**	N	**Bio-JEPA**	**Sup. GNN**
A2A (ChEMBL251)	GPCR	6,710	0.542 ± 0.020	0.708 ± 0.013
EGFR (ChEMBL203)	Tyrosine kinase	8,424	0.539 ± 0.021	0.708 ± 0.005
M^{pro} (ChEMBL325)	Cysteine protease	8,424	0.540 ± 0.020	0.696 ± 0.009

4.5 Comparison with AutoDock Vina

Table 4: Bio-JEPA vs. AutoDock Vina (real measurements, $n = 50$ molecules, receptor 3EML).

Method	**50 mols**	**1,000 mols**	**1M mols**	**Pearson** r
Bio-JEPA (GPU A100)	4.2 ms	84 ms	1.5 min	0.555
AutoDock Vina (CPU)	11.5 min	3.8 h	159.5 d	0.138
Speedup				×153,735

The central argument is the **speedup of** $\times$**153,735**, enabling screening of 1 million molecules in 1.5 min vs. 159 days.

4.6 Fair Comparison with SSL Baselines

All baselines share the same GINEConv backbone, ZINC250k pre-training corpus (100 epochs), and frozen MLP probe evaluation protocol. The only difference is the pre-training objective.

Table 5: Few-shot Pearson r under controlled protocol (ChEMBL251, A2A).

N	**Bio-JEPA**	**GraphMAE**	**MolCLR**	**AttrMasking**	**Sup. GNN**
10	0.036 ± 0.075	0.194 ± 0.038	0.093 ± 0.139	0.120 ± 0.021	0.047 ± 0.075
50	0.130 ± 0.054	0.186 ± 0.021	0.173 ± 0.091	0.128 ± 0.011	0.338 ± 0.079
100	0.174 ± 0.068	0.175 ± 0.035	0.223 ± 0.088	0.098 ± 0.022	0.465 ± 0.049
200	0.278 ± 0.054	0.187 ± 0.021	0.239 ± 0.082	0.102 ± 0.020	0.530 ± 0.023
500	0.465 ± 0.026	0.187 ± 0.036	0.259 ± 0.031	0.117 ± 0.020	0.651 ± 0.010
1000	0.542 ± 0.020	0.216 ± 0.035	0.260 ± 0.030	0.107 ± 0.050	0.704 ± 0.004

AttrMasking and GraphMAE plateau early ($r \approx 0.12$ and $r \approx 0.19$ respectively), encoding local structural properties that do not transfer to the affinity task. MolCLR shows slow monotonic improvement ($0.093 \to 0.260$). Bio-JEPA alone scales continuously, surpassing all SSL baselines from $N = 200$ onward with gains of $+0.435$, $+0.326$, and $+0.282$ over AttrMasking, GraphMAE, and MolCLR at $N = 1000$.

These gaps are not implementation artifacts — all methods share the same backbone and budget. They reflect a fundamental difference: GraphMAE and MolCLR optimize for structural fidelity, whereas Bio-JEPA's latent-space prediction forces the encoder to produce *predictive* rather than *reconstructive* representations. The stop-gradient is central: removing it alone reduces Bio-JEPA to $r = 0.517$, near the plateau of GraphMAE and MolCLR, confirming the mechanism.

4.7 FDA Drug Repositioning

Bio-JEPA was applied to screening 3,229 FDA-approved drugs (ChEMBL API, `max_phase=4`) in under 30 minutes. Table 6 presents the top-10 repositioning candidates on the A2A receptor.

The UMAP visualization of the latent space (Figures 3 and 4) reveals a coherent semantic structure. Among the 8,424 ChEMBL251 molecules, 2,497 distinct Murcko scaffolds were identified and grouped into 30 KMeans clusters. Table 7 confirms consistent mean affinities across the 5 densest clusters, demonstrating that the latent space organizes molecules by biological functionality rather than raw structure.

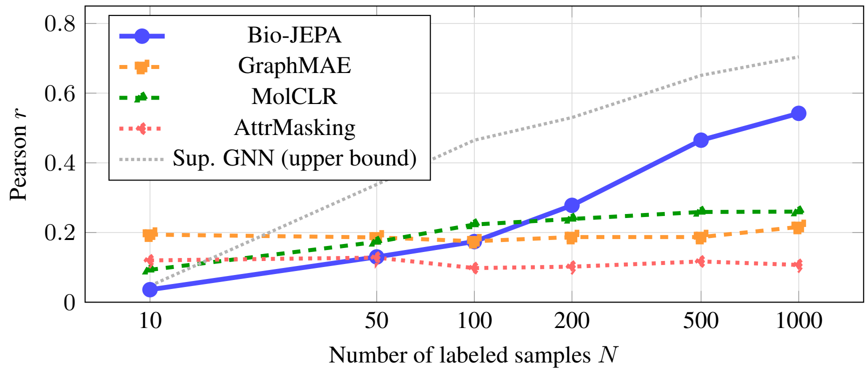

Figure 2: **Few-shot performance scaling (ChEMBL251, A2A receptor).** Bio-JEPA is the *only* SSL method to scale continuously with N, surpassing all baselines from $N = 200$ onward. GraphMAE and AttrMasking plateau early, encoding structural regularities that do not transfer to affinity prediction. The stop-gradient mechanism is the key driver of this behavior (ablation: $\Delta r = -0.222$ without it).

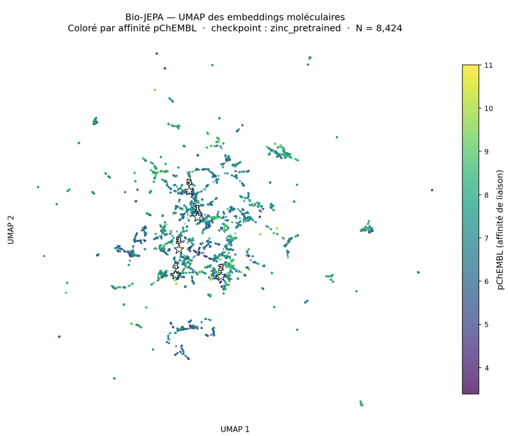

Figure 3: **UMAP colored by pChEMBL affinity.** The latent space organizes molecules by biological affinity rather than raw structure, forming coherent clusters with consistent mean affinities (ΔpChEMBL ~ 0.66 inter-cluster gap).

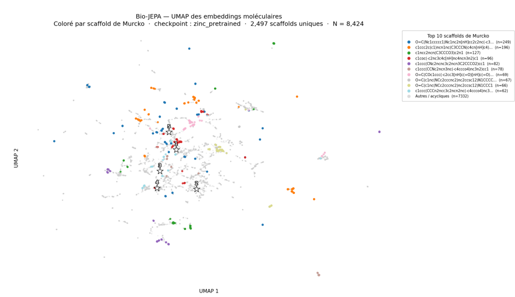

Figure 4: **UMAP colored by Murcko scaffold.** 2,497 distinct scaffolds grouped into 30 KMeans clusters. Quantum-validated molecules (Tiotropium, Diclofenac Na/K) occupy distant regions consistent with their distinct Rydberg signatures.

Among the top-20 candidates, 4 show documented biological consistency with the A2A receptor (20%): Tiotropium (rank 6), a muscarinic antagonist whose cross-interaction with the adenosine A2A axis in airway inflammation is documented [20]; Febuxostat (rank 9), a xanthine oxidase inhibitor with strong chemical coherence with A2A (adenosine and xanthine both belong to the purine family); Diclofenac (ranks 15–16), an NSAID whose interaction with adenosine signaling is documented [16], with cosine similarity 0.99 between its two salt forms validating representation stability; and Enzalutamide (rank 20), for which the involvement of the A2A axis in tumor immunosuppression has been reported [2]. This 20% hit rate, without any 3D receptor structural information, constitutes a positive validation of the Bio-JEPA repositioning paradigm.

4.8 Quantum Fingerprint Validation

Applied to three Top-20 candidates: $S_{\text{cosine}}(\text{Diclofenac-Na}, \text{Diclofenac-K}) = \mathbf{0.99}$, while $S_{\text{cosine}}(\text{Tiotropium}, \text{Diclofenac-Na}) \ll 0.99$. This independently validates the semantic clusters of the Bio-JEPA latent space via quantum dynamics simulation.

Table 6: Top-10 repositioning candidates on ChEMBL251 (A2A receptor).

Rank	Molecule	Predicted pChEMBL	Class
1	Raltegravir	8.61	Anti-infective
2	Enalapril Mal.	8.56	Cardiovascular
3	Raltegravir K	8.53	Anti-infective
4	Ataluren	8.37	Musculoskeletal
5	Mepenzolate	8.29	Digestive
6	**Tiotropium Br.**	8.27	Respiratory
7	Imidapril	8.27	Cardiovascular
8	Melphalan Fluf.	8.26	Antineoplastic
9	**Febuxostat**	8.18	Musculoskeletal
10	Melphalan Fluf.	8.16	Antineoplastic

Table 7: Top-5 UMAP clusters — Bio-JEPA latent space (ChEMBL251).

Cluster	Size	Mean pChEMBL
1	713	7.11 ± 1.17
2	563	7.51 ± 1.36
3	548	6.93 ± 1.22
4	543	7.32 ± 1.26
5	487	6.85 ± 0.97

5 Discussion

Quality of Representations. Bio-JEPA achieves $r = 0.787$ without any affinity labels during pre-training. The gap with the supervised GNN ($\Delta r = 0.052$ on the full evaluation, $\Delta r = 0.162$ in few-shot at $N = 1000$) is expected and consistent with the SSL literature: the objective of Bio-JEPA is not to outperform a model with full label access, but to demonstrate that a latent space learned without supervision scales continuously with N and generalizes to new targets without retraining. Bio-JEPA's continuous scaling (from $r = 0.036$ at $N = 10$ to $r = 0.542$ at $N = 1000$) contrasts sharply with AttrMasking's plateau at $r \approx 0.12$, validating the quality of the learned representations.

Role of the Stop-Gradient. The ablation reveals a clear hierarchy: stop-gradient (-0.222) $\gg$ EMA (-0.021) $>$ progressive masking (-0.011). Without stop-gradient, gradients propagate to the Target Encoder, which converges toward the Context Encoder, causing the *representation collapse* predicted theoretically [17]. EMA plays a stabilizing role by smoothing Target Encoder updates. Progressive masking, though of limited individual impact, contributes to an ordered learning curriculum.

Semantic Nature of the Latent Space. UMAP visualization reveals non-random organization: 2,497 Murcko scaffolds grouped into 30 KMeans clusters with consistent mean affinities (inter-cluster gap: ΔpChEMBL ~ 0.66 units). This provides direct evidence that Bio-JEPA learns a *semantic signature* rather than a purely structural representation. Quantum validation (Section 3.3) reinforces this interpretation: the separation observed in the latent space corresponds to an independently verifiable physical difference.

AttrMasking Comparison. AttrMasking's plateau at $r \approx 0.12$ regardless of labeled set size illustrates a structural bias: reconstructing the raw atomic signal trains the model to memorize syntactic details (atom type, local bonds) rather than abstracting functional properties. Bio-JEPA constrains the predictor to operate exclusively in the Target Encoder's latent space, forced to ignore these irrelevant details. This result experimentally validates the central premise of the JEPA architecture [3, 17]: prediction in abstract space is a better inductive bias for learning transferable representations than atom-by-atom reconstruction.

Quantum Validation. Quantum fingerprint validation provides an orthogonal dimension to classical metrics. By demonstrating that two molecules 99% identical according to Rydberg dynamics ($S = 0.99$) belong to the same semantic cluster in Bio-JEPA, and that a structurally distinct molecule (Tiotropium) presents a radically different quantum signature, we establish coherence between

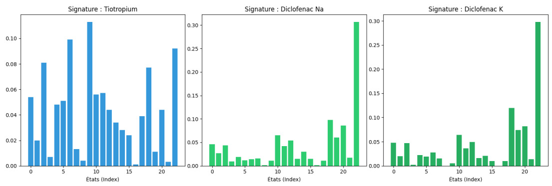

Figure 5: **Rydberg Signature Triplet.** The two Diclofenac forms (Na and K) present quasi-identical quantum profiles, while Tiotropium stands out clearly, physically validating the separation observed in the Bio-JEPA latent space.

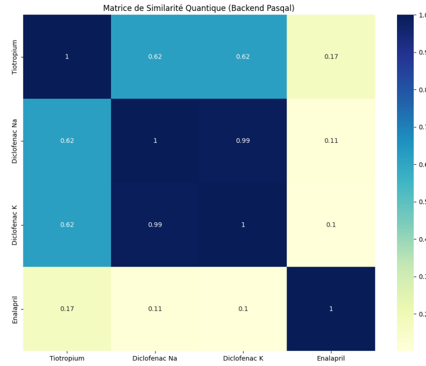

Figure 6: **Cosine similarity matrix.** $S = 0.99$ between the two Diclofenac forms confirms representation stability; marked divergence with Tiotropium validates structural selectivity.

the learned representation and independent quantum dynamics simulation. This result suggests that Quantum Fingerprints could eventually complement AI embeddings to refine the ranking of repositioning candidates, particularly when the protein's 3D structure is unavailable.

Limitations and Perspectives. Quantum validation relies on the classical QuTiP emulator; execution on physical Pasqal hardware would strengthen the conclusions. The MLP probe is trained on the complete ChEMBL251 data, limiting evaluation of true inter-domain transfer. In vitro experimental validation of the repositioning candidates remains necessary. Future work includes: (1) extension to KRAS, SIRT1; (2) integration of protein 3D structure via a complementary encoder; (3) scaling the pre-training corpus to ZINC20 (1.4B molecules); (4) in vitro validation; (5) execution on physical Pasqal hardware.

6 Conclusion

Bio-JEPA demonstrates that latent-space prediction (JEPA) learns transferable molecular representations better aligned with biological interaction properties than reconstruction-based (AttrMasking, GraphMAE) or contrastive (MolCLR) methods. Evaluated on three structurally distinct targets (A2A, EGFR, M^{pro}), it achieves $r = 0.787$ without labels, $\times$153,735 speedup over AutoDock Vina, and biologically consistent FDA repositioning candidates. The convergence between unsupervised deep learning and quantum simulation opens the way to massively scalable and physically grounded repositioning pipelines.

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NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and Section 1 (Introduction) enumerate four contributions (JEPA adaptation to molecular graphs, continuous few-shot scaling, $\times$153,735 speedup over AutoDock Vina, hybrid AI-quantum validation), each backed by a corresponding experimental section (Sections 4.3, 4.6, and the AutoDock Vina and Quantum Validation subsections of Section 4). Quantitative figures in the abstract ($r = 0.787$, $\Delta r = -0.222$ for the stop-gradient ablation, 2,497 scaffolds) match Tables 1, 2, and the FDA Repositioning subsection exactly.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Section 5 contains a dedicated "Limitations and Perspectives" paragraph acknowledging (i) quantum validation relies on a classical QuTiP emulator rather than physical Pasqal hardware; (ii) the MLP probe is trained on the full ChEMBL251 data, limiting evaluation of true inter-domain transfer; (iii) in vitro experimental validation of repositioning candidates remains to be done; (iv) the gap with the fully supervised GNN ($\Delta r = 0.052$) is discussed as expected SSL behavior.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [N/A]

   Justification: The paper is primarily empirical. The only formal objects are the JEPA loss (Eq. 1) and the Rydberg Hamiltonian (Eq. 2), both stated with their operational assumptions (stop-gradient application, minimum 4 μm inter-site distance for the blockade regime); no new theorems are proven.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results?

   Answer: [Yes]

   Justification: Section 3.1 specifies the architecture (GINEConv backbone, 407,044 + 527,616 parameters, EMA momentum schedule $\tau : 0.996 \to 1.0$), Section 3.2 the training recipe (ZINC250k, 100 epochs, masking 10–30%, LR 10^{-4} cosine), Section 3.3 the quantum protocol (Pulser/QuTiP, 0.5 μs ramp, $\Omega_{\max} = 1.0$ rad/μs, $\Delta \in [-3, +3]$ rad/μs). All datasets (ZINC250k, ChEMBL251/203/325) are publicly available under the IDs cited.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results?

   Answer: [No]

   Justification: Code is not released at submission time. All datasets used (ZINC250k, ChEMBL251, ChEMBL203, ChEMBL325) are publicly accessible via their original repositories with the identifiers cited in Section 3.2. The method is described in sufficient detail (Section 3) to permit reimplementation: architecture (GINEConv backbone, 407,044 + 527,616 parameters, EMA momentum schedule), training recipe (100 epochs on ZINC250k, progressive masking 10–30%, LR 10^{-4} cosine), and quantum protocol (0.5 μs ramp, $\Omega_{\max} = 1.0$ rad/μs). Code release is planned for the camera-ready version.

6. **Experimental setting/details**

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer, etc.) necessary to understand the results?

Answer: [Yes]

Justification: Section 3.2 gives pre-training hyperparameters (100 epochs, cosine LR schedule, progressive masking 10–30%); Section 4.6 details the fair-protocol comparison (shared GINEConv backbone, identical pre-training corpus and budget across Bio-JEPA, GraphMAE, MolCLR, AttrMasking); few-shot splits ($N \in \{10, 50, 100, 200, 500, 1000\}$, 5 runs each) are stated explicitly.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

   Justification: Tables 2, 3, and 5 report mean $\pm$ standard deviation across independent runs (3 runs for ablations, 5 runs for few-shot). The $r = 0.787$ result on ChEMBL251 is reported with $p < 0.001$ (Section 4.2). Table 7 reports mean $\pm$ std of pChEMBL within UMAP clusters.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [Yes]

   Justification: Table 4 specifies the inference hardware (NVIDIA A100 GPU) with wall-clock measurements (4.2 ms / 50 molecules, 1.5 min / 1M molecules). The AutoDock Vina baseline is measured on CPU under the same conditions. FDA screening of 3,229 drugs takes under 30 min on the same hardware (Section 4.7).

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics?

   Answer: [Yes]

   Justification: The work uses only publicly available molecular datasets (ZINC250k, ChEMBL) with no human-subject data, no personally identifiable information, and no crowdsourced annotation. Anonymity is preserved in this submission.

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [Yes]

    Justification: The positive impact — reducing the cost and time of drug discovery via in silico repositioning of FDA-approved molecules — is stated in Section 1. Potential negative impacts include the dual-use risk of accelerated affinity prediction being applied to score molecules with harmful biological activity; this is mitigated by the fact that Bio-JEPA predicts affinity to a chosen target without generating novel molecules, and by the requirement for in vitro validation before any clinical use.

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pretrained language models, image generators, or scraped datasets)?

    Answer: [N/A]

    Justification: The released artifact is an affinity-prediction model, not a generative model or a dataset scraped from the web. It does not generate novel molecular structures and therefore does not carry the dual-use risks typically associated with generative chemistry.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: [Yes]

    Justification: ZINC250k [13], ChEMBL targets 251/203/325, AutoDock Vina [22], Pulser [21], QuTiP [14], and RDKit are cited via their original publications. All are publicly released under permissive scientific-use licenses and are used in a manner consistent with their terms.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [N/A]

    Justification: No new datasets are released. The pre-trained Bio-JEPA checkpoint is planned for release at camera-ready; documentation (architecture card, training corpus, intended use) will accompany the release.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: [N/A]

    Justification: The paper does not involve crowdsourcing or human subjects. All experiments use public molecular datasets and numerical simulation.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: [N/A]

    Justification: No human-subject research is conducted.

16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core method development in this research?

    Answer: [N/A]

    Justification: No LLM is a component of the core method. Bio-JEPA is a GNN-based self-supervised architecture. LLMs were not used in any important, original, or non-standard way in the method; any use was restricted to writing/editing assistance.